Can Artificial Intelligence beat an efficient market?

Netflix uses artificial intelligence (AI) to analyse viewing habits and recommend programmes based on patterns in consumer behaviour. This raises an interesting question: if machine learning can predict what people might watch, could similar techniques predict movements in financial markets?

Financial institutions increasingly use algorithmic trading systems to analyse market data and execute transactions automatically. Some employ machine learning to identify patterns in historical prices, trading volumes and other financial information. Unlike conventional trading strategies, which follow predefined rules, machine-learning models can identify complex statistical relationships that might otherwise be overlooked.

However, the ability to identify patterns does not necessarily mean an investor can consistently profit from them. Eugene Fama's Efficient Market Hypothesis (EMH) argues that financial asset prices reflect available information[1]. Under this theory, consistently achieving returns above those justified by risk is difficult because new information is rapidly incorporated into market prices. If an algorithm identifies an undervalued asset, other investors may discover the same opportunity, causing its price to adjust and eliminating the potential advantage.

This creates an interesting contradiction. As machine-learning systems become more sophisticated, they may improve market efficiency by identifying and exploiting pricing discrepancies more quickly. Yet the more effectively these systems operate, the harder it becomes for individual investors to find profitable opportunities. Andrew Lo offers a different perspective, suggesting that market efficiency is not necessarily constant[2]. His Adaptive Markets Hypothesis views financial markets as evolving environments in which investors compete, learn and adapt their strategies. Under this interpretation, an algorithm might perform successfully under particular market conditions but become less effective as competitors adopt similar approaches or economic circumstances change.

[1] Fama, ‘Efficient Capital Markets: A Review of Theory and Empirical Work’, 383.
[2] Lo, ‘Efficient Markets Hypothesis’, 14–15.

There are also important limitations to algorithmic prediction. Financial markets are influenced by unexpected political events, changes in monetary policy and shifts in investor confidence. Historical patterns may not remain reliable when these conditions change. Furthermore, transaction costs and the risks associated with inaccurate predictions can undermine apparently profitable strategies. The comparison with Netflix therefore has limits. Predicting a viewer's preferences is different from predicting market prices, partly because traders respond strategically to one another. A successful financial prediction can itself influence the market it attempts to anticipate. Ultimately, AI may not disprove the Efficient Market Hypothesis. Instead, its growing use raises a more interesting possibility: that technological competition continuously changes how efficiently markets process information.

References

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